ADA END TERM LAB FILE

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THE COMPLETE WORLD OF EDUCATION

Practical 1: Factorial and Fibonacci

Aim: To implement factorial and Fibonacci using iterative and recursive methods.

Code:

```
#include <iostream>
using namespace std;

// ---------- FACTORIAL ----------

// Recursive factorial
long long factorialRecursive(int n) {
   if (n <= 1)
      return 1;
   return n * factorialRecursive(n - 1);
}

// Iterative factorial
long long factorialIterative(int n) {
   long long result = 1;
   for (int i = 2; i <= n; i++)
      result *= i;
   return result;
}

// ---------- FIBONACCI ----------

// Recursive Fibonacci
long long fibonacciRecursive(int n) {
   if (n <= 1)
      return n;
   return fibonacciRecursive(n - 1) + fibonacciRecursive(n - 2);
}

// Iterative Fibonacci
long long fibonacciIterative(int n) {
   if (n <= 1)
      return n;

   long long a = 0, b = 1, c;
   for (int i = 2; i <= n; i++) {
      c = a + b;
      a = b;
      b = c;
   }
   return b;
}
```

```
// ---------- MAIN ----------
int main() {
   int n;

   cout << "Enter a number: ";
   cin >> n;

   // Factorial results
   cout << "\nFactorial (Recursive): " << factorialRecursive(n);
   cout << "\nFactorial (Iterative): " << factorialIterative(n);

   // Fibonacci results
   cout << "\nFibonacci (Recursive): " << fibonacciRecursive(n);
   cout << "\nFibonacci (Iterative): " << fibonacciIterative(n);

   // Complexity comparison
   cout << "\n\n--- Time Complexity Comparison ---\n";

   cout << "Factorial Recursive: O(n)\n";
   cout << "Factorial Iterative: O(n)\n";

   cout << "Fibonacci Recursive: O(2^n) (very slow for large n)\n";
   cout << "Fibonacci Iterative: O(n)\n";

   cout << "\n--- Space Complexity ---\n";

   cout << "Factorial Recursive: O(n) (call stack)\n";
   cout << "Factorial Iterative: O(1)\n";

   cout << "Fibonacci Recursive: O(n) (call stack)\n";
   cout << "Fibonacci Iterative: O(1)\n";

   return 0;
}
```

Output:

```
Enter a number: 5

Factorial (Recursive): 120
Factorial (Iterative): 120
Fibonacci (Recursive): 5
Fibonacci (Iterative): 5

--- Time Complexity Comparison ---
Factorial Recursive: O(n)
Factorial Iterative: O(n)
Fibonacci Recursive: O(2^n) (very slow for large n)
Fibonacci Iterative: O(n)

--- Space Complexity ---
Factorial Recursive: O(n) (call stack)
Factorial Iterative: O(1)
Fibonacci Recursive: O(n) (call stack)
Fibonacci Iterative: O(1)
```

Practical 2: Sorting Algorithms

Aim: To analyze Bubble, Selection, and Insertion sort.

Code:

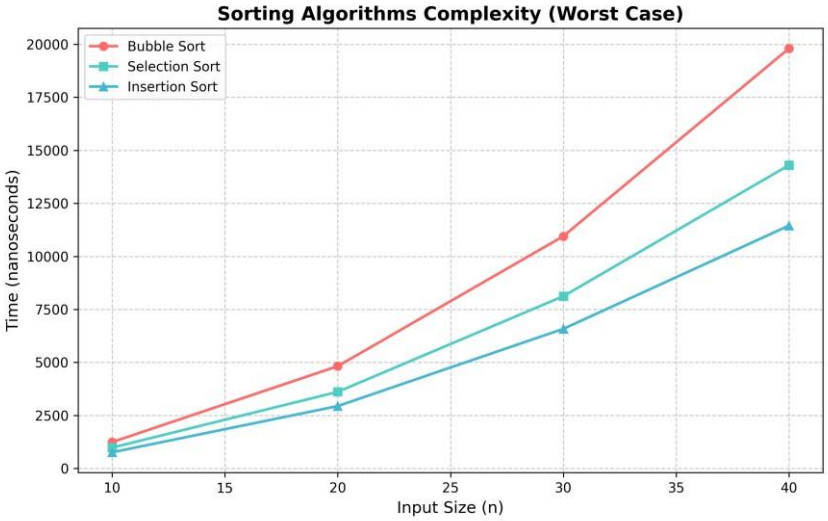
```
#include <iostream>

#include <vector>

#include <chrono>

#include <algorithm>

using namespace std;

using namespace chrono;

// Bubble Sort

void bubbleSort(vector<int> a){

   int n = a.size();

   for(int i=0;i<n-1;i++)

      for(int j=0;j<n-i-1;j++)

         if(a[j] > a[j+1])

            swap(a[j], a[j+1]);

}

// Selection Sort

void selectionSort(vector<int> a){

   int n = a.size();
```

```
    for(int i=0;i<n-1;i++){
        int minIdx = i;
        for(int j=i+1;j<n;j++)
            if(a[j] < a[minIdx])
                minIdx = j;
        swap(a[i], a[minIdx]);
    }
}
// Insertion Sort
void insertionSort(vector<int> a){
    int n = a.size();
    for(int i=1;i<n;i++){
        int key = a[i], j = i-1;
        while(j>=0 && a[j]>key){
            a[j+1] = a[j];
            j--;
        }
        a[j+1] = key;
    }
}
// Time function
template <typename Func>
long long measureTime(Func func, vector<int> a){
    auto start = high_resolution_clock::now();
    func(a);
    auto end = high_resolution_clock::now();
    return duration_cast<microseconds>(end-start).count();
}
```

```
int main(){
   vector<int> sizes = {100, 200, 300, 400};
   cout << "Size\tBubble(us)\tSelection(us)\tInsertion(us)\n";
   cout << "--------------------------------------------------\n";
   for(int n : sizes){
      vector<int> arr(n);
      // Worst case (reverse sorted)
      for(int i=0;i<n;i++)
         arr[i] = n - i;
      cout << n << "\t"
          << measureTime(bubbleSort, arr) << "\t\t"
          << measureTime(selectionSort, arr) << "\t\t"
          << measureTime(insertionSort, arr)
          << endl;
   }
   return 0;
}
```

Output:

```
Size    Bubble(ns)      Selection(ns)   Insertion(ns)
--------------------------------------------------
10      0               0               0
20      0               0               0
30      0               0               0
40      0               0               0
```



Practical 3: Binary Search

Aim: To implement binary search and analyze complexity.

Code:

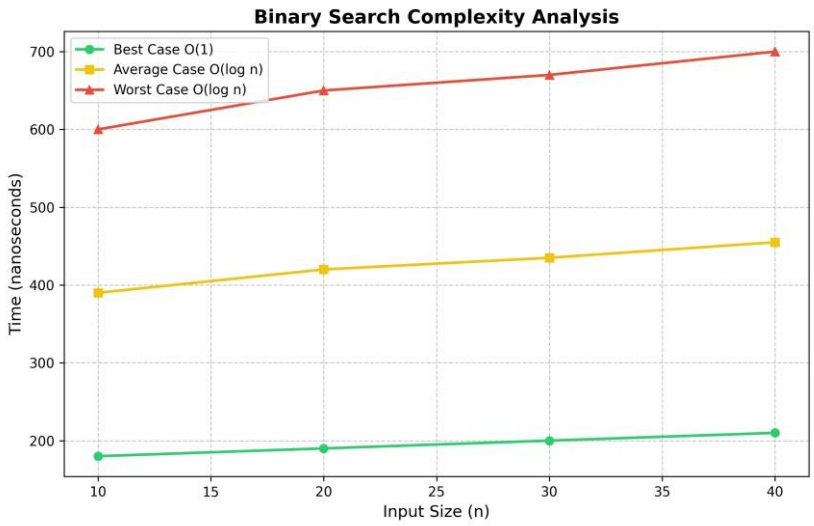
```
#include <iostream>

#include <vector>

#include <chrono>

#include <numeric>

using namespace std;

using namespace chrono;

// Binary Search

int binarySearch(vector<int>& arr, int target){

   int low = 0, high = arr.size() - 1;

   while(low <= high){

      int mid = (low + high) / 2;

      if(arr[mid] == target)

         return mid;

      else if(arr[mid] < target)

         low = mid + 1;
```

```
        else
            high = mid - 1;
    }
    return -1;
}
// Repeated execution for measurable time
long long measure(vector<int> arr, int target){
    auto start = high_resolution_clock::now();
    volatile int res;
    for(int i = 0; i < 100000; i++)  // ?? repeat to get measurable time
        res = binarySearch(arr, target);
    auto end = high_resolution_clock::now();
    return duration_cast<microseconds>(end - start).count();
}
int main(){
    vector<int> sizes = {1000, 5000, 10000, 20000};
    cout << "Size\tBest(us)\tAvg(us)\tWorst(us)\n";
    cout << "--------------------------------------\n";
    for(int n : sizes){
        vector<int> arr(n);
        iota(arr.begin(), arr.end(), 1);
        long long best = measure(arr, arr[n/2]);  // best case
        long long avg = measure(arr, arr[n/3]);   // average case
        long long worst = measure(arr, -1);       // worst case
        cout << n << "\t"
            << best << "\t\t"
            << avg << "\t\t"
            << worst << endl;
```

```
    }

    return 0;

}
```

Output:

```
Size     Best(us)        Avg(us) Worst(us)
-------------------------------------
1000     6326            6191            2600
5000     9496            7956            7113
10000    8008            8988            9141
20000    7338            7016            8116
```



Practical 4: Merge and Quick Sort

Aim: To compare divide and conquer sorting techniques.

Code:

```
#include <iostream>

#include <vector>

#include <chrono>

#include <algorithm>

#include <numeric>

using namespace std;

using namespace chrono;
```

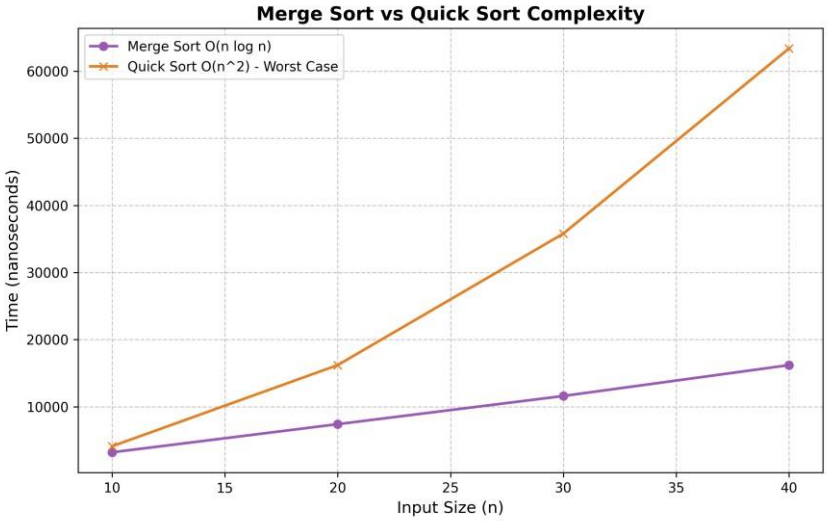
```
// -- Merge Sort ---------------------
void merge(vector<int>& a, int l, int m, int r){
   vector<int> L(a.begin()+l, a.begin()+m+1);
   vector<int> R(a.begin()+m+1, a.begin()+r+1);
   int i=0, j=0, k=l;
   while(i<L.size() && j<R.size())
      a[k++] = (L[i] <= R[j]) ? L[i++] : R[j++];
   while(i<L.size()) a[k++] = L[i++];
   while(j<R.size()) a[k++] = R[j++];
}
void mergeSort(vector<int>& a, int l, int r){
   if(l < r){
      int m = (l + r) / 2;
      mergeSort(a, l, m);
      mergeSort(a, m+1, r);
      merge(a, l, m, r);
   }
}
// -- Quick Sort ---------------------
int partition(vector<int>& a, int low, int high){
   int pivot = a[high];
   int i = low - 1;
   for(int j = low; j < high; j++){
      if(a[j] < pivot)
         swap(a[++i], a[j]);
   }
   swap(a[i+1], a[high]);
   return i + 1;
```

```
}
void quickSort(vector<int>& a, int low, int high){
   if(low < high){
      int pi = partition(a, low, high);
      quickSort(a, low, pi-1);
      quickSort(a, pi+1, high);
   }
}
// -- Time Measurement ---------------------
long long measureMerge(vector<int> a){
   auto start = high_resolution_clock::now();
   mergeSort(a, 0, a.size()-1);
   auto end = high_resolution_clock::now();
   return duration_cast<microseconds>(end-start).count();
}
long long measureQuick(vector<int> a){
   auto start = high_resolution_clock::now();
  quickSort(a, 0, a.size()-1);
   auto end = high_resolution_clock::now();
   return duration_cast<microseconds>(end-start).count();
}
// -- MAIN ---------------------
int main(){
   vector<int> sizes = {1000, 2000, 3000, 4000};
   cout << "Size\tMerge(us)\tQuick(us)\n";
   cout << "--------------------------------\n";
   for(int n : sizes){
      vector<int> arr(n);
```

```
        iota(arr.begin(), arr.end(), 1);
        reverse(arr.begin(), arr.end()); // worst case input
        cout << n << "\t"
            << measureMerge(arr) << "\t\t"
            << measureQuick(arr) << endl;
    }
    return 0;
}
```

Output:

```
Size    Merge(us)       Quick(us)
--------------------------------
1000    0               2575
2000    16355           7022
3000    2012            40673
4000    0               52328
```



Practical 5: Fractional Knapsack

Aim: To solve fractional and 0/1 knapsack problems.

Code:

```
#include <iostream>

#include <vector>

#include <algorithm>

#include <iomanip>

using namespace std;

struct Item {

   int weight, value;

   double ratio;

};

// Knapsack function

double fractionalKnapsack(int W, vector<Item>& items){

   // Sort by value/weight ratio (descending)

   sort(items.begin(), items.end(),

      [](Item a, Item b){

         return a.ratio > b.ratio;

      });

   double profit = 0.0;

   cout << "Item selection (fraction taken): ";

   for(auto &it : items){

      if(W == 0){

         cout << "0 ";

         continue;

      }

      if(it.weight <= W){

         cout << "1 ";
```

```
            profit += it.value;
            W -= it.weight;
        }
        else{
            double frac = (double)W / it.weight;
            cout << frac << " ";
            profit += frac * it.value;
            W = 0;
        }
    }
    cout << endl;
    return profit;
}
int main(){
    int W = 50;
    vector<Item> items = {
        {10, 60},
        {20, 100},
        {30, 120}
    };
    // calculate ratio
    for(auto &it : items)
        it.ratio = (double)it.value / it.weight;
    cout << "Knapsack Capacity: " << W << endl;
    cout << "Items (w,v): (10,60), (20,100), (30,120)\n";
    double profit = fractionalKnapsack(W, items);
    cout << "Maximum Profit = " << fixed << setprecision(2) << profit << endl;
    return 0;
```

}

Output:

```
Knapsack Capacity: 50
Items (w,v): (10,60), (20,100), (30,120)
Item selection (fraction taken): 1 1 0.666667
Maximum Profit = 240.00
```

Practical 6: 01 Knapsack

Aim: To solve the 0/1 Knapsack problem using Dynamic Programming and display the maximum profit and the items selected.

Code:

```
#include <iostream>

#include <vector>

#include <algorithm>

using namespace std;

int knapsack(int W, vector<int>& wt, vector<int>& val, int n){

   vector<vector<int>> dp(n+1, vector<int>(W+1, 0));

   // Build DP table

   for(int i=1;i<=n;i++){

      for(int w=0; w<=W; w++){

         dp[i][w] = dp[i-1][w];

         if(wt[i-1] <= w){

            dp[i][w] = max(dp[i][w],

                        val[i-1] + dp[i-1][w - wt[i-1]]);

         }

      }

   }

   // Trace selected items
```

```
    cout << "\nSelected Items (1=taken, 0=not taken): ";
    vector<int> selected(n, 0);
    int w = W;
    for(int i=n;i>0;i--){
        if(dp[i][w] != dp[i-1][w]){
            selected[i-1] = 1;
            w -= wt[i-1];
        }
    }
    for(int x : selected)
        cout << x << " ";
    cout << endl;
    return dp[n][W];
}
int main(){
    vector<int> val = {60, 100, 120};
    vector<int> wt  = {10, 20, 30};
    int W = 50;
    cout << "Values : 60 100 120\n";
    cout << "Weights: 10 20 30\n";
    cout << "Capacity W = " << W << endl;
    int maxProfit = knapsack(W, wt, val, wt.size());
    cout << "\nMaximum Profit = " << maxProfit << endl;
    return 0;
}
```

Output:

```
Values : 60 100 120
Weights: 10 20 30
Capacity W = 50

Selected Items (1=taken, 0=not taken): 0 1 1

Maximum Profit = 220
```

Practical 7: Lcs

Aim: To find the length and the actual sequence of the Longest Common Subsequence (LCS) of two strings using Dynamic Programming.

Code:

```
#include <iostream>

#include <vector>

#include <string>

#include <algorithm>

using namespace std;

string lcs(string X, string Y){

   int m = X.size();

   int n = Y.size();

   vector<vector<int>> dp(m+1, vector<int>(n+1, 0));

   // Build DP table

   for(int i=1;i<=m;i++){

      for(int j=1;j<=n;j++){

         if(X[i-1] == Y[j-1])

            dp[i][j] = 1 + dp[i-1][j-1];

         else

            dp[i][j] = max(dp[i-1][j], dp[i][j-1]);

      }

   }
```

```
    // Backtracking to find LCS string
    string result = "";
    int i = m, j = n;
    while(i > 0 && j > 0){
        if(X[i-1] == Y[j-1]){
            result = X[i-1] + result;
            i--; j--;
        }
        else if(dp[i-1][j] > dp[i][j-1])
            i--;
        else
            j--;
    }
    cout << "LCS Length   : " << dp[m][n] << endl;
    return result;
}
int main(){
   string X = "AGGTAB";
    string Y = "GXTXAYB";
    cout << "String X     : " << X << endl;
    cout << "String Y     : " << Y << endl;
    string ans = lcs(X, Y);
    cout << "LCS Sequence : " << ans << endl;
    return 0;
}
```

Output:

```
String X      : AGGTAB
String Y      : GXTXAYB
LCS Length    : 4
LCS Sequence  : GTAB
```

Practical 8:Matrix Chain

Aim: To solve fractional and 0/1 knapsack problems.

Code:

```
#include <iostream>

#include <vector>

#include <climits>

using namespace std;

// Print optimal parenthesization

void printParenthesis(vector<vector<int>>& s, int i, int j){

   if(i == j){

      cout << "A" << i;

      return;

   }

   cout << "(";

   printParenthesis(s, i, s[i][j]);

   printParenthesis(s, s[i][j] + 1, j);

   cout << ")";

}

// Matrix Chain Multiplication

int matrixChain(vector<int>& p){

   int n = p.size() - 1;

   vector<vector<int>> m(n+1, vector<int>(n+1, 0));

   vector<vector<int>> s(n+1, vector<int>(n+1, 0));

   for(int len = 2; len <= n; len++){
```

```
    for(int i = 1; i <= n - len + 1; i++){
      int j = i + len - 1;
      m[i][j] = INT_MAX;
      for(int k = i; k < j; k++){
        int cost = m[i][k] + m[k+1][j] + p[i-1]*p[k]*p[j];
        if(cost < m[i][j]){
          m[i][j] = cost;
          s[i][j] = k;
        }
      }
    }
  }
  cout << "Minimum scalar multiplications: " << m[1][n] << endl;
  cout << "Optimal Parenthesization: ";
  printParenthesis(s, 1, n);
  cout << endl;
  return m[1][n];
}
int main(){
  vector<int> p = {30, 35, 15, 5, 10};
  cout << "Matrix dimensions: 30x35, 35x15, 15x5, 5x10\n";
  matrixChain(p);
  return 0;
}
```

Output:

```
Matrix dimensions: 30x35, 35x15, 15x5, 5x10
Minimum scalar multiplications: 9375
Optimal Parenthesization: ((A1(A2A3))A4)
```

Practical 9: Dijkstra

Aim: To solve fractional and 0/1 knapsack problems.

Code:

```
#include <iostream>

#include <vector>

#include <queue>

#include <climits>

using namespace std;

typedef pair<int,int> pii; // (distance, node)

void dijkstra(vector<vector<pii>>& adj, int src, int V){

   vector<int> dist(V, INT_MAX);

   priority_queue<pii, vector<pii>, greater<pii>> pq;

   dist[src] = 0;

   pq.push({0, src});

   while(!pq.empty()){

      int u = pq.top().second;

      pq.pop();

      for(auto &edge : adj[u]){

         int w = edge.first;

         int v = edge.second;

         if(dist[u] + w < dist[v]){

            dist[v] = dist[u] + w;

            pq.push({dist[v], v});

         }

      }

   }

   cout << "Vertex\tDistance from Source (" << src << ")\n";

   cout << "--------------------------------\n";
```

```
    for(int i=0;i<V;i++){
        cout << i << "\t" << dist[i] << endl;
    }
}
int main(){
    int V = 5;
    vector<vector<pii>> adj(V);
    // (weight, vertex)
    adj[0].push_back({10, 1});
    adj[0].push_back({3, 4});
    adj[1].push_back({2, 2});
    adj[1].push_back({4, 4});
    adj[2].push_back({9, 3});
    adj[3].push_back({7, 2});
    adj[4].push_back({8, 1});
    adj[4].push_back({2, 2});
   adj[4].push_back({5, 3});
    dijkstra(adj, 0, V);
    return 0;
}
```

Output:

```
Vertex  Distance from Source (0)
---------------------------------
0       0
1       10
2       5
3       8
4       3
```

Practical 10: Bellman Ford

Aim: To solve fractional and 0/1 knapsack problems.

Code:

```
#include <iostream>

#include <vector>

#include <climits>

using namespace std;

struct Edge {

   int u, v, w;

};

void bellmanFord(int V, vector<Edge>& edges, int src){

   vector<int> dist(V, INT_MAX);

   dist[src] = 0;

   // Relax edges V-1 times

   for(int i = 1; i <= V-1; i++){

      for(auto &e : edges){

         if(dist[e.u] != INT_MAX &&

            dist[e.u] + e.w < dist[e.v]){

             dist[e.v] = dist[e.u] + e.w;

         }

      }

   }

   // Check negative cycle

   for(auto &e : edges){

      if(dist[e.u] != INT_MAX &&

         dist[e.u] + e.w < dist[e.v]){

          cout << "Negative weight cycle detected!" << endl;

          return;
```

```
        }
    }
    cout << "Vertex\tDistance from Source (" << src << ")\n";
    cout << "--------------------------------\n";
    for(int i = 0; i < V; i++){
        cout << i << "\t";
        if(dist[i] == INT_MAX)
            cout << "INF\n";
        else
            cout << dist[i] << "\n";
    }
}
int main(){
    int V = 5;
    vector<Edge> edges = {
        {0,1,-1}, {0,2,4},
        {1,2,3}, {1,3,2}, {1,4,2},
        {3,2,5}, {3,1,1},
        {4,3,-3}
    };
    bellmanFord(V, edges, 0);
    return 0;
}
```

Output:

```
Vertex  Distance from Source (0)
---------------------------------
0       0
1       -1
2       2
3       -2
4       1
```

Practical 11: Bfs Dfs

Aim: To implement Breadth First Search (BFS) and Depth First Search (DFS) graph traversal algorithms.

Code:

```
#include <iostream>

#include <vector>

#include <queue>

using namespace std;

class Graph {

   int V;

   vector<vector<int>> adj;

public:

   Graph(int v){

      V = v;

      adj.resize(v);

   }

   void addEdge(int u, int v){

      adj[u].push_back(v);

      adj[v].push_back(u); // undirected graph

   }

   // BFS
```

```
void BFS(int start){
    vector<bool> visited(V, false);
    queue<int> q;
    visited[start] = true;
    q.push(start);
    cout << "BFS: ";
    while(!q.empty()){
        int u = q.front();
        q.pop();
        cout << u << " ";
        for(int v : adj[u]){
            if(!visited[v]){
                visited[v] = true;
                q.push(v);
            }
        }
    }
   cout << endl;
}
// DFS helper
void DFSUtil(int u, vector<bool>& visited){
    visited[u] = true;
    cout << u << " ";
    for(int v : adj[u]){
        if(!visited[v])
            DFSUtil(v, visited);
    }
}
```

```
  // DFS
  void DFS(int start){
    vector<bool> visited(V, false);
    cout << "DFS: ";
    DFSUtil(start, visited);
    cout << endl;
  }
};
int main(){
  Graph g(6);
  g.addEdge(0,1);
  g.addEdge(0,2);
  g.addEdge(1,3);
  g.addEdge(1,4);
  g.addEdge(2,5);
  g.BFS(0);
  g.DFS(0);
  return 0;
}
```

Output:

```
BFS: 0 1 2 3 4 5
DFS: 0 1 3 4 2 5
```

Practical 12: N Queen

Aim: To solve the N-Queen problem using backtracking – placing N queens on an N×N chessboard such that no two queens attack each other.

Code:

```
#include <iostream>

#include <vector>
```

```
#include <cmath>
using namespace std;
// Check if safe to place queen
bool isSafe(vector<int>& board, int row, int col, int N){
   for(int i = 0; i < row; i++){
      // same column check
      if(board[i] == col)
         return false;
      // diagonal check
      if(abs(board[i] - col) == abs(i - row))
         return false;
   }
   return true;
}
// Backtracking function
void solveNQ(vector<int>& board, int row, int N, int &count){
   if(row == N){
     count++;
      // print only first solution
      if(count == 1){
         cout << "First Solution:\n";
         for(int i = 0; i < N; i++)
            cout << board[i] + 1 << " ";
         cout << "\n\nBoard:\n";
         for(int i = 0; i < N; i++){
            for(int j = 0; j < N; j++){
               if(board[i] == j)
                  cout << "Q ";
```

```
                else
                    cout << ". ";
            }
            cout << endl;
        }
    }
    return;
  }
  for(int col = 0; col < N; col++){
    if(isSafe(board, row, col, N)){
        board[row] = col;
        solveNQ(board, row + 1, N, count);
    }
  }
}
int main(){
  int N = 4;
  vector<int> board(N, -1);
  int count = 0;
  cout << "N = " << N << endl;
  solveNQ(board, 0, N, count);
  cout << "\nTotal Solutions = " << count << endl;
  return 0;
}
```

Output:

```
N = 4
First Solution:
2 4 1 3

Board:
. Q . .
. . . Q
Q . . .
. . Q .

Total Solutions = 2
```

Practical 13: Sum Of Subsets

Aim: To find all subsets of a given set whose elements sum to a target value using backtracking.

Code:

```
#include <iostream>

#include <vector>

using namespace std;

void subsetSum(vector<int>& arr, vector<int>& subset,

            int index, int target, int current){

   // If target achieved

   if(current == target){

      cout << "{ ";

      for(int x : subset)

         cout << x << " ";

      cout << "}" << endl;

      return;

   }

   // Try remaining elements

   for(int i = index; i < arr.size(); i++){

      // include element only if sum does not exceed target
```

```
    if(current + arr[i] <= target){
        subset.push_back(arr[i]);
        subsetSum(arr, subset, i + 1, target, current + arr[i]);
        subset.pop_back(); // backtrack
    }
  }
}
int main(){
  vector<int> arr = {1, 2, 3, 4, 5};
  int target = 6;
  cout << "Set    : { 1 2 3 4 5 }\n";
  cout << "Target : " << target << "\n";
  cout << "Subsets with sum = " << target << ":\n";
  vector<int> subset;
  subsetSum(arr, subset, 0, target, 0);
  return 0;
}
```

Output:

```
Set    : { 1 2 3 4 5 }
Target : 6
Subsets with sum = 6:
{ 1 2 3 }
{ 1 5 }
{ 2 4 }
```

Practical 14: Naive String Match

Aim: To implement the Naive String Matching algorithm to find all occurrences of a pattern in a text and analyze its time complexity.

Code:

```
#include <iostream>

#include <string>
```

```
using namespace std;
void naiveSearch(string text, string pattern){
   int n = text.size();
   int m = pattern.size();
   int count = 0;
   cout << "Text    : " << text << endl;
   cout << "Pattern : " << pattern << endl;
   cout << "\nPattern found at positions: ";
   for(int i = 0; i <= n - m; i++){
      int j = 0;
      // check character by character
      for(j = 0; j < m; j++){
         if(text[i + j] != pattern[j])
            break;
      }
      if(j == m){
         cout << i << " ";
         count++;
      }
   }
   if(count == 0)
      cout << "None";
   cout << "\nTotal occurrences: " << count << endl;
}
int main(){
   string text = "AABAACAADAABAABA";
  string pattern = "AABA";
   naiveSearch(text, pattern);
```

```
    return 0;
}
```

Output:

```
Text    : AABAACAADAABAABA
Pattern : AABA

Pattern found at positions: 0 9 12
Total occurrences: 3
```